Graphs 1 — Gradient · Straight-line graphs · Conversion graphs

You will need squared paper (or a printed grid) for the questions that say *draw*. Marks are shown in brackets. ★ = harder question.

A · GRADIENT

1 Find the gradient of the line joining each pair of points. [4]

a $A(1,\ 2)$ and $B(4,\ 11)$

b $C(-3,\ 4)$ and $D(5,\ -2)$

2 A wheelchair ramp rises 1.2 m over a horizontal distance of 8 m. [3]

a Work out the gradient of the ramp.

b Building rules say a ramp should have a gradient of at most $\frac{1}{12}$. How long (horizontally) must a ramp with the same rise be to meet the rule?

B · STRAIGHT-LINE GRAPHS

3 For the line $y = 3x - 2$: [5]

a Write down its gradient and its y-intercept.

b Copy and complete the table of values for $x = -1,\ 0,\ 1,\ 2,\ 3$.

c Draw the graph for $-1 \le x \le 3$.

d Use your graph (or the equation) to find where the line crosses the x-axis.

4 A line has equation $2x + 5y = 10$. [4]

a Find the points where it crosses the two axes.

b Draw the line.

c Work out its gradient.

5 Here are four lines: $y = 2x + 1$, $2y = x + 6$, $y - 2x = 5$, $x + 2y = 4$. [3]
Which two lines are parallel? Explain how you know.

C · REAL-LIFE AND CONVERSION GRAPHS

6 A taxi charges a fixed €3 plus €1.50 per kilometre, so the cost C (euros) of a journey of d km is $C = 3 + 1.5d$. [6]

a Draw the graph of C against d for $0 \le d \le 20$.

b What is the gradient of the graph, and what does it represent?

c Use your graph to find how far you can travel for €21.

7 5 miles is approximately 8 kilometres. [5]

a Draw a conversion graph from miles to kilometres for 0 to 50 miles.

b Use it to convert 35 miles to kilometres.

c Use it to convert 100 km to miles.

8 A candle is 24 cm tall and burns down at a steady 2 cm per hour. [5]

a Write an equation for the height h cm of the candle t hours after it is lit.

b Draw the graph of h against t for $0 \le t \le 12$.

c Use your graph to find when the candle is a quarter of its original height.

Total: 35 marks

Check: gradient = rise ÷ run, with the sign · three well-spaced points for a straight line · first-named variable on the vertical axis · label axes with units.